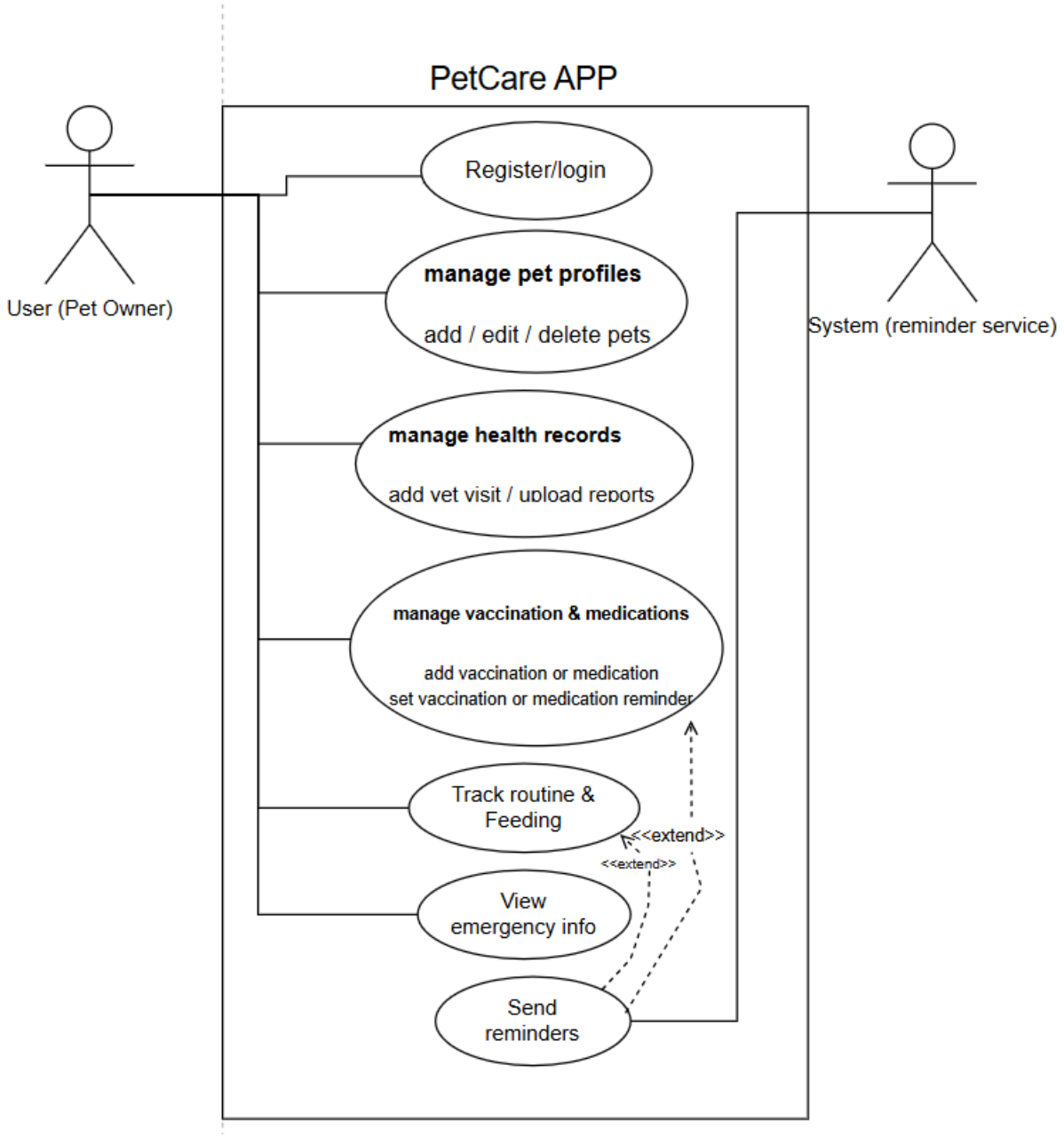
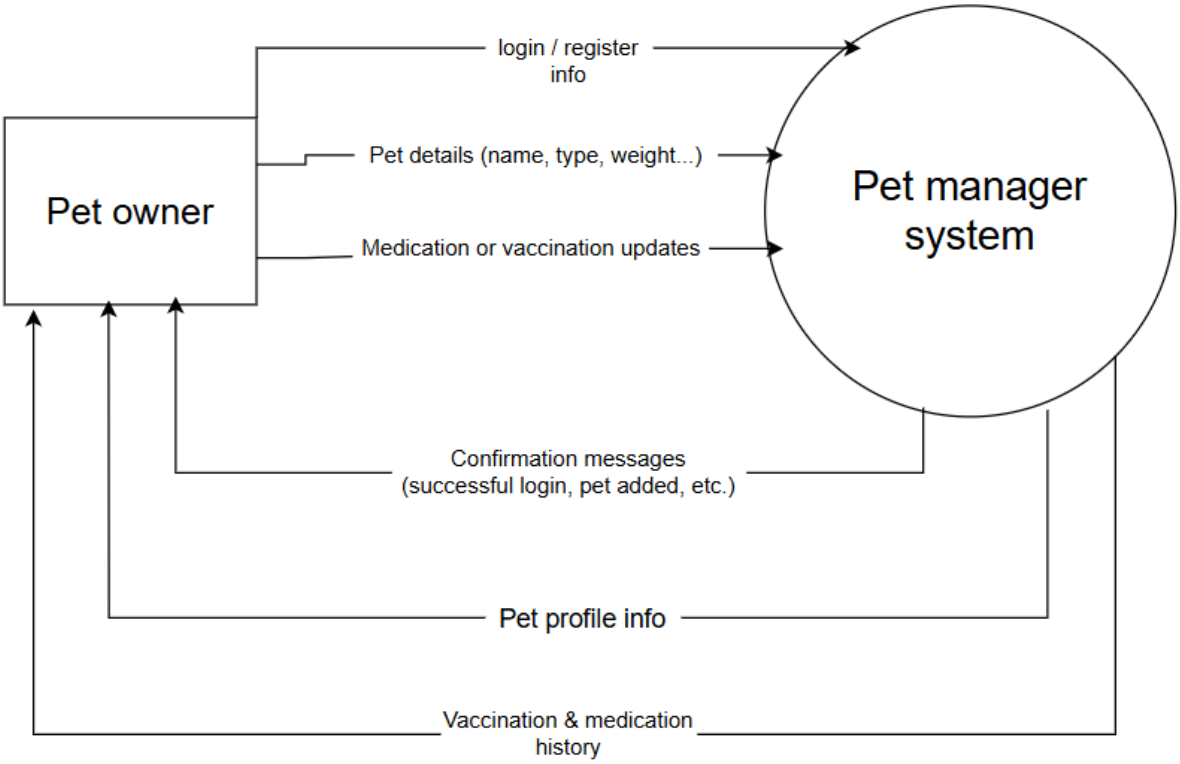
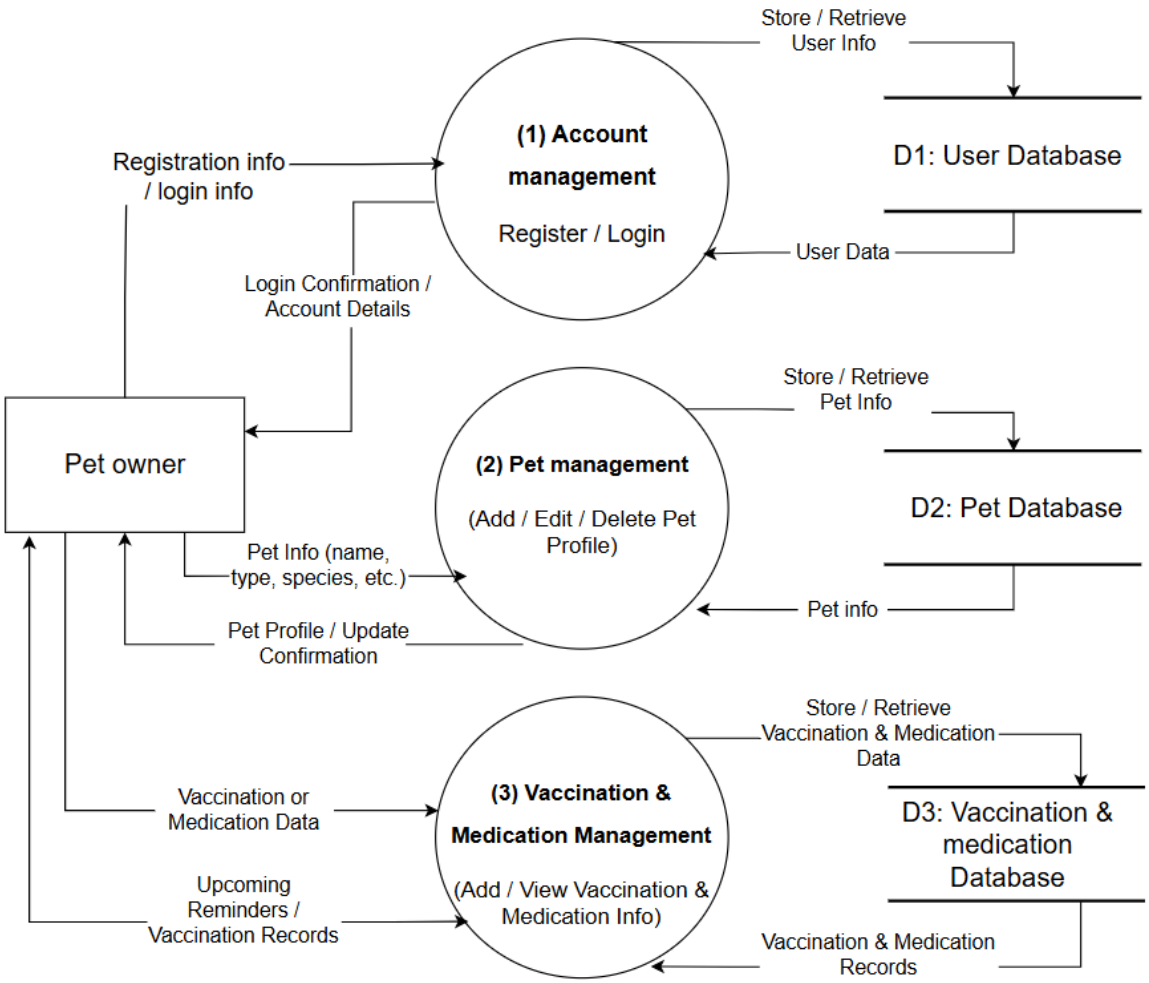
Use case


PetCare APP
User (Pet Owner)
System (reminder service)
Register/login
manage pet profiles
add / edit / delete pets
manage health records
add vet visit / upload reports
manage vaccination & medications
add vaccination or medication
set vaccination or medication reminder
Track routine & Feeding
View emergency info
Send reminders
<<extend>>
<<extend>>

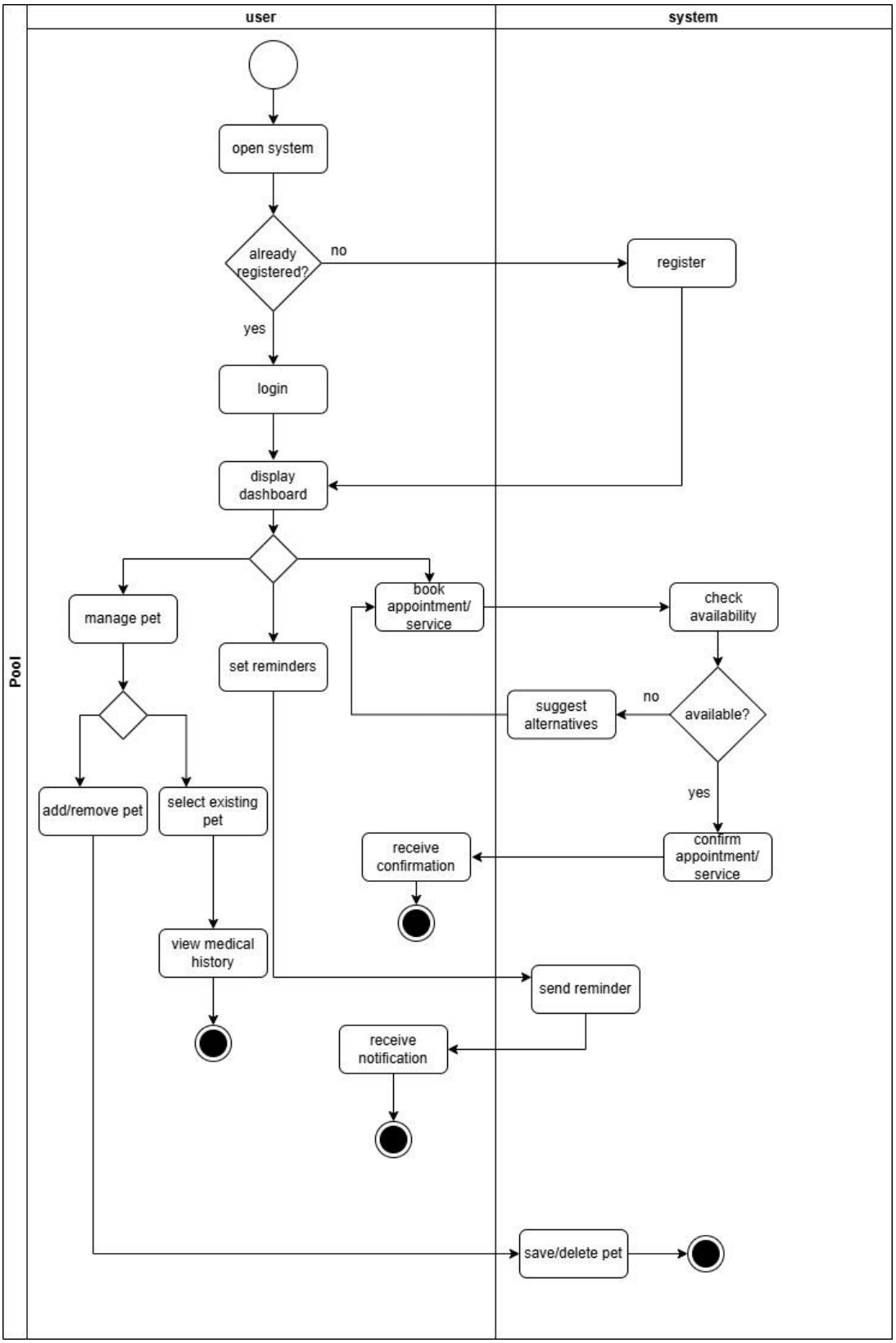
DFD lvl 0


Pet owner
login / register info
Pet details (name, type, weight...)
Medication or vaccination updates
Pet manager system
Confirmation messages (successful login, pet added, etc.)
Pet profile info
Vaccination & medication history

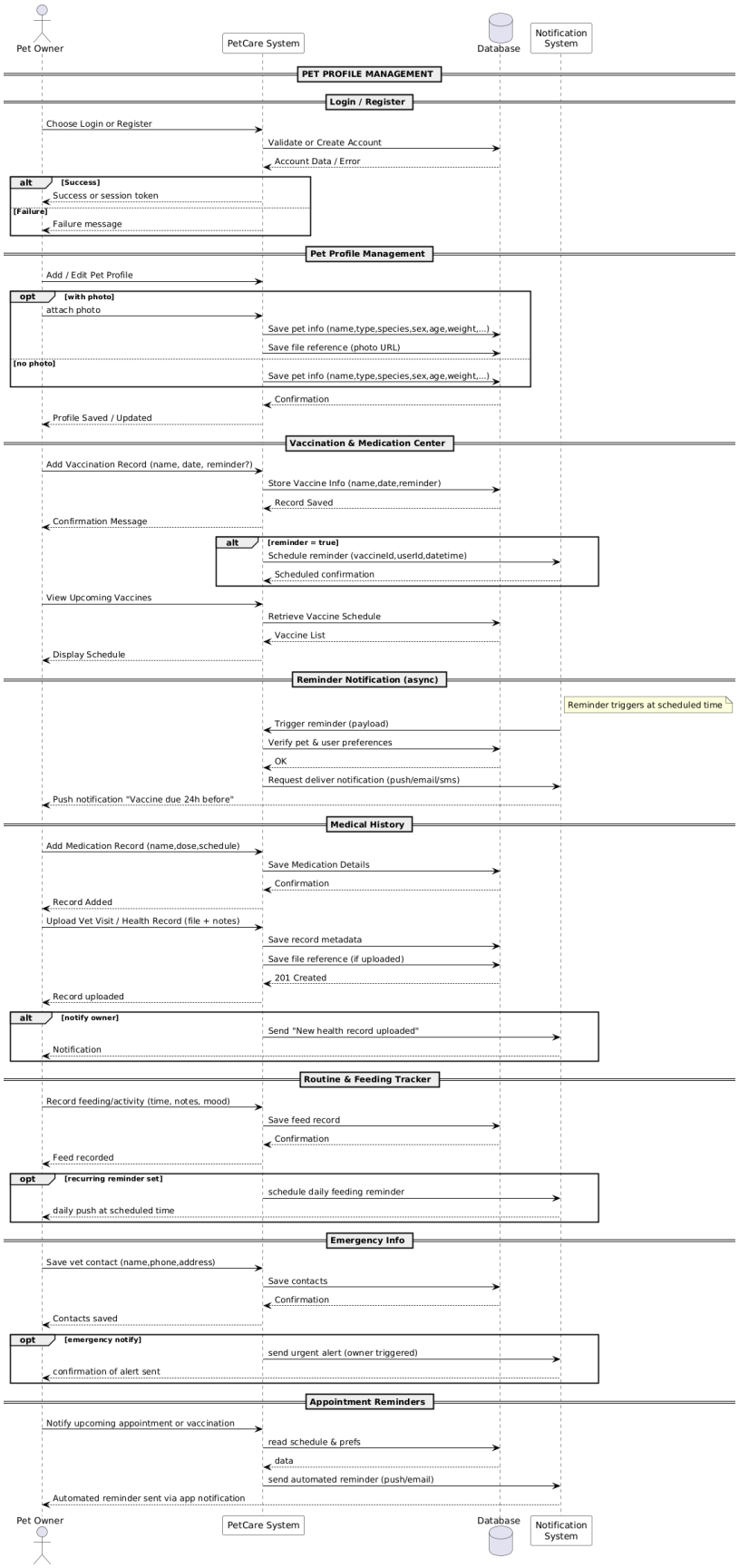
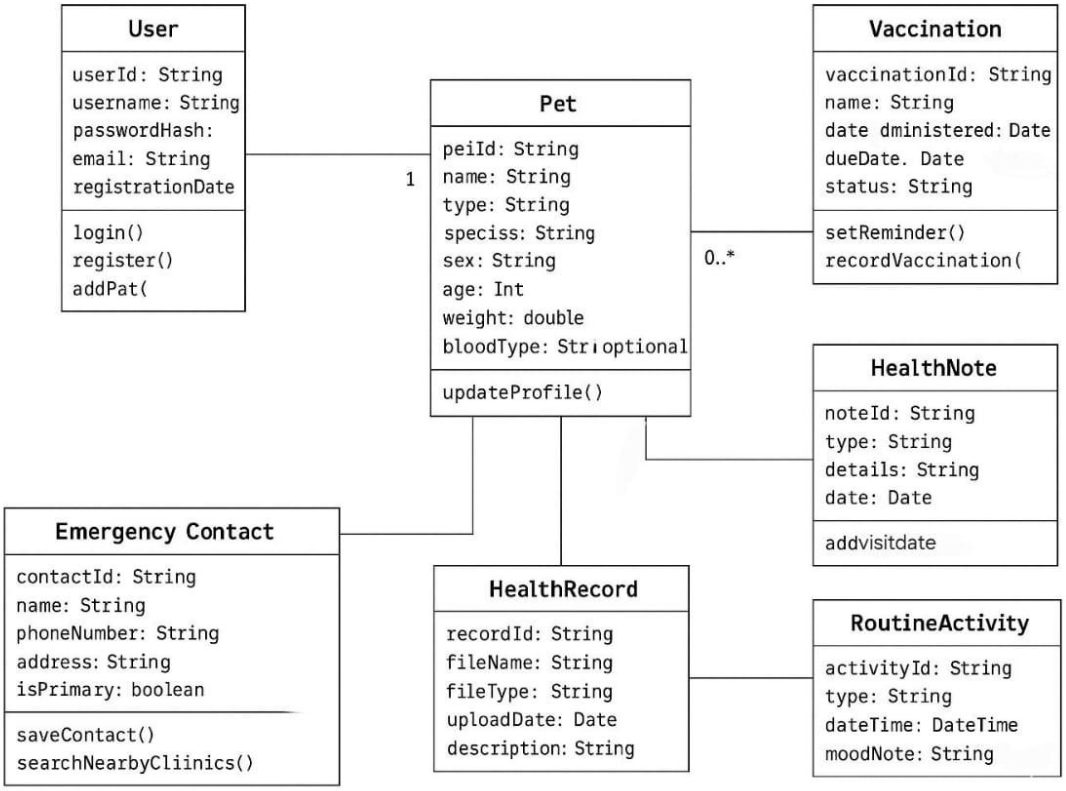
DFD lvl 1


Store / Retrieve User Info
(1) Account management
Register / Login
D1: User Database
User Data
Registration info / login info
Login Confirmation / Account Details
Pet owner
Store / Retrieve Pet Info
(2) Pet management
(Add / Edit / Delete Pet Profile)
D2: Pet Database
Pet Info (name, type, species, etc.)
Pet info
Pet Profile / Update Confirmation
Store / Retrieve Vaccination & Medication Data
(3) Vaccination & Medication Management
(Add / View Vaccination & Medication Info)
D3: Vaccination & medication Database
Vaccination or Medication Data
Upcoming Reminders / Vaccination Records
Vaccination & Medication Records

Activity diagram



Sequence diagram


Pet Owner
PetCare System
Database
Notification System
PET PROFILE MANAGEMENT
Login / Register
Choose Login or Register
Validate or Create Account
Account Data / Error
alt
[Success]
Success or session token
[Failure]
Failure message
Pet Profile Management
Add / Edit Pet Profile
opt
[with photo]
attach photo
Save pet info (name,type,species,sex,age,weight,...)
Save file reference (photo URL)
[no photo]
Save pet info (name,type,species,sex,age,weight,...)
Confirmation
Profile Saved / Updated
Vaccination & Medication Center
Add Vaccination Record (name, date, reminder?)
Store Vaccine Info (name,date,reminder)
Record Saved
Confirmation Message
alt
[reminder = true]
Schedule reminder (vaccineId,userId,datetime)
Scheduled confirmation
View Upcoming Vaccines
Retrieve Vaccine Schedule
Vaccine List
Display Schedule
Reminder Notification (async)
Reminder triggers at scheduled time
Trigger reminder (payload)
Verify pet & user preferences
OK
Request deliver notification (push/email/sms)
Push notification "Vaccine due 24h before"
Medical History
Add Medication Record (name,dose,schedule)
Save Medication Details
Confirmation
Record Added
Upload Vet Visit / Health Record (file + notes)
Save record metadata
Save file reference (if uploaded)
201 Created
Record uploaded
alt
[notify owner]
Send "New health record uploaded"
Notification
Routine & Feeding Tracker
Record feeding/activity (time, notes, mood)
Save feed record
Confirmation
Feed recorded
opt
[recurring reminder set]
schedule daily feeding reminder
daily push at scheduled time
Emergency Info
Save vet contact (name,phone,address)
Save contacts
Confirmation
Contacts saved
opt
[emergency notify]
send urgent alert (owner triggered)
confirmation of alert sent
Appointment Reminders
Notify upcoming appointment or vaccination
read schedule & prefs
data
send automated reminder (push/email)
Automated reminder sent via app notification
Pet Owner
PetCare System
Database
Notification System

Class diagram